

The Graded Gap: Forced Condensation, the Stability of Democracy, and Where Hierarchy Must Come From

Harald Rønning

Abstract. Step B of the summit road executes the gap program on the graded structure, and begins with a precision point the variational analysis enforced: the marriage of chiralities must be the *field-level* pairing $\varphi_L \varphi_R$ — a relevant operator — because the current-current version is marginal in two dimensions and can only renormalize the velocity, verified exactly (symplectic frequencies $k\sqrt{1 - \lambda^2}$ to six digits, no gap). The series' own current algebra cannot mass itself; mass lives at the field level, in the infrared — *the same home as the memory* — precisely where the state-stratum reading placed it. Three results follow, each machine-verified. **(1) Forced condensation.** In the per-axis (two-dimensional) setting the variational energy diverges at the massless point — the infrared wall of $\langle \varphi^2 \rangle$ — falls to a unique interior minimum, and rises toward the ultraviolet ($+1.570 \rightarrow +0.504$ at $m^* = 1.53 \rightarrow +2.700$); the gap identity $m_*^2 = 3g Q(m_*)$ emerges at the minimum without being imposed (0.4-1.1% across $g = 0.3, 1, 3$). *The vacuum cannot afford masslessness*: condensation is not one branch among two but forced, and selection is by minimization — the summit's mechanism-and-branch question closed for the model, with the series' oldest theme returning inverted: the infrared, which every previous paper tamed, here *compels* the condensate. **(2) Democracy is dynamically stable.** Under a τ -symmetric quartic the two-channel minimum sits exactly on the diagonal (mass ratio 1.0000): hierarchy cannot arise from democratic interactions at mean field — the dynamical completion of the generation paper's kinematic democracy. **(3) Channel asymmetry splits, monotonically.** Asymmetric couplings split the minimum (g -ratio $1.5 \rightarrow$ mass ratio $1.16; 3.0 \rightarrow 1.52$). The framework owns exactly one *intrinsic* source of such asymmetry: the rigid grading — the three layer channels are algebraically inequivalent in kind, so vertices built through the algebra are channel-asymmetric by structure, not by choice. The chain ***rigid grading $\rightarrow$ asymmetric effective couplings $\rightarrow$ split masses*** is thereby established at both ends, with the middle — the actual channel weights from the algebra's associator-sensitive vertex structure — named as the next computation.

1. The precision point: mass lives with the memory

The mass paper labeled the marriage $m \int j_L j_R$; the variational analysis of this note forces the correction. A current-current coupling is dimension two — marginal — and its exact effect is a velocity renormalization: the coupled per- k block has symplectic frequencies $k\sqrt{1 - \lambda^2}$ (verified to six digits), gapless as $k \rightarrow 0$. The structure that gaps is the *relevant*, field-level pairing $\varphi_L \varphi_R$, whose 2×2 dispersion problem is exactly the one the mass paper diagonalized and verified. The correction deepens the story rather than disturbing it: the field is the object whose zero mode is the memory, the object the infrared condition governs — **mass and memory share a home**, and the condensate is an infrared structure of the state, exactly as the non-kinematic placement required. (The mass paper's operator

label is flagged for the consolidation pass; every verified statement there — dispersion, correlation length, Klein anatomy — stands.)

2. Forced condensation: the vacuum cannot afford masslessness

With the trial family $|m\rangle$ of married vacua and a quartic interaction treated variationally (Gaussian factorization), the energy density is

$$\varepsilon(m) = K(m) + \frac{3g}{4} Q(m)^2, \quad K(m) = \frac{1}{4\pi} \int_0^\Lambda \left(\omega + \frac{k^2}{\omega}\right) dk, \quad Q(m) = \frac{1}{2\pi} \int_0^\Lambda \frac{dk}{\omega},$$

and the two-dimensional infrared does the deciding: $Q(m) \sim \frac{1}{2\pi} \ln(2\Lambda/m)$ diverges as $m \rightarrow 0$, so the massless point costs *infinite* energy. Verified at $g = 1, \Lambda = 100$: ε falls from +1.5701 at $m = 0.02$ to the unique minimum +0.5045 at $m_* = 1.534$, then rises to +2.6996 at $m = 8$. At the minimum the gap identity appears as the stationarity condition it is: m_*^2 against $3gQ(m_*)$ gives 2.352 vs 2.326, and at $g = 0.3$ and 3.0 : 0.774 vs 0.777 and 6.337 vs 6.267 — grid-resolution agreement, nothing imposed. The summit's first question — what selects the branch — is closed for the model: there is only one branch the vacuum can occupy, and minimization finds it. The series' infrared theme returns with the sign reversed: the infrared divergence that every earlier paper had to tame is here the *engine* — masslessness is the one thing the two-dimensional vacuum cannot pay for.

3. The stability of democracy

The hierarchy question in its sharpest mean-field form: does a τ -symmetric interaction ever prefer asymmetric masses? With two channels and the symmetric quartic $(g/4)(\varphi_1^2 + \varphi_2^2)^2$, the two-dimensional minimization lands *exactly* on the diagonal: $(m_1, m_2) = (1.7459, 1.7459)$, ratio 1.0000. **Democratic interactions preserve democracy** — the dynamical completion of the generation-grading paper's kinematic verdict. Hierarchy is not an instability waiting in symmetric dynamics; at this level it simply does not happen.

4. Where hierarchy must come from

Break the couplings and the masses follow, monotonically: $g_{11}/g_{22} = 1.5$ gives mass ratio 1.163; 3.0 gives 1.524. So the question becomes: what, inside this framework, makes couplings unequal? Not the state's whim — the framework owns exactly one *intrinsic* source: **the rigid grading**. The three layer channels through which the splittings act are algebraically inequivalent in kind — two octonionic, one non-composition, with no automorphism relating any pair — so effective vertices built through the algebra's own structure constants and associators are channel-asymmetric *by theorem*, not by choice. The chain is now established at both ends: the algebra pre-orders the channels (rigid-grading paper), and asymmetric channels split the spectrum (this paper, monotonically). The middle link — computing the actual channel weights from the associator-sensitive vertex structure — is the named next construction; beyond-mean-field corrections belong to the perturbative-algebraic road. For the *generation* direction the ledger stands as before: cross-generation structure enters through the charged channels of the Klein anatomy, and inter-

generation hierarchy requires spontaneous $\mathbb{Z}_3$ breaking — which §3 shows democratic dynamics will not supply, sharpening that question rather than closing it.

5. The summit's ledger, updated

(S1) Mechanism and selection: closed in-model — condensation forced by the infrared wall, the state selected by minimization, the gap identity emergent. **(S2) Structure:** a no-go half — democratic interactions preserve democracy, kinematically (generation paper) and now dynamically — and a constructive half: the rigid grading is the in-framework breaking resource, with the channel-weight computation as the open middle link; generation hierarchy remains conditioned on $\mathbb{Z}_3$ breaking that symmetric mean field will not provide. **(S3) The scale:** standing — the gap transmutes coupling and cutoff into mass, and the tower's dimension-three μ remains the candidate seed. The summit has narrowed from a face to two marked routes.

6. Scope

Five items. (i) All results are variational/mean-field with Gaussian factorization; normal-ordering conventions are absorbed into the subtracted functional. (ii) The forced-condensation wall is specific to the two-dimensional (per-axis) infrared; the assembled $3+1$ version has a different infrared and is flagged, not extrapolated. (iii) The channel-weight computation from the algebra's associators is the named next step. (iv) Beyond-mean-field structure belongs to the perturbative-algebraic scaffold. (v) The precision note of §1 is logged for the consolidation pass over the mass paper.

References

- Companion papers: *Mass as Marriage: The Chirality Pairing, the Unit Condensate, and the Klein Anatomy of Mass Matrices*; *The Rigid Grading: Layer Inequivalence, the Cross-Split Marriage, and the Algebra's Pre-Order on Mass Channels*; *The Massive Interior: Ultraviolet Safety, Causality with Tails, and What Survives Mass*; *The Generation Grading: The Klein Selection Rule, Democratic Gauge Universality, and the Cyclic Generation Symmetry*; *The Soft Sector: Memory as a Superselection Label and the Harmonic-Number Structure of the Zero-Mode Kernel*.

Created in collaboration with Claude (Anthropic). All conclusions and any errors remain the author's responsibility.